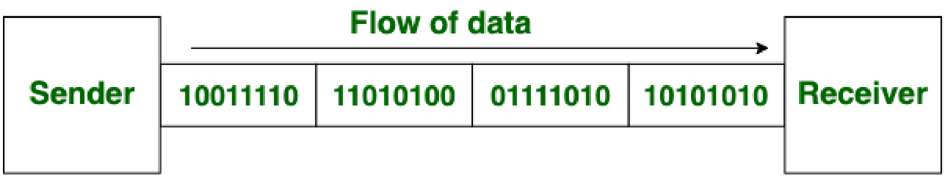
1.3 Asynchronous and Synchronous transmission , XDSL , Introduction to Wired and Wireless LAN

Asynchronous and Synchronous Transmission:

Synchronous Transmission: In Synchronous Transmission, data is sent in form of blocks or frames. This transmission is the full-duplex type. Between sender and receiver, synchronization is compulsory. In Synchronous transmission, There is no gap present between data. It is more efficient and more reliable than asynchronous transmission to transfer a large amount of data.

Example:

- Chat Rooms
- Telephonic Conversations
- Video Conferencing

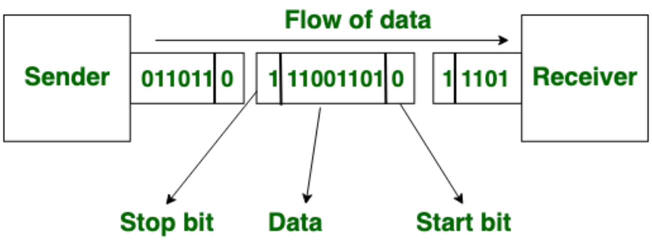


Synchronous Transmission

Asynchronous Transmission: In Asynchronous Transmission, data is sent in form of byte or character. This transmission is the half-duplex type transmission. In this transmission start bits and stop bits are added with data. It does not require synchronization.

Example:

- Email
- Forums
- Letters



Asynchronous Transmission

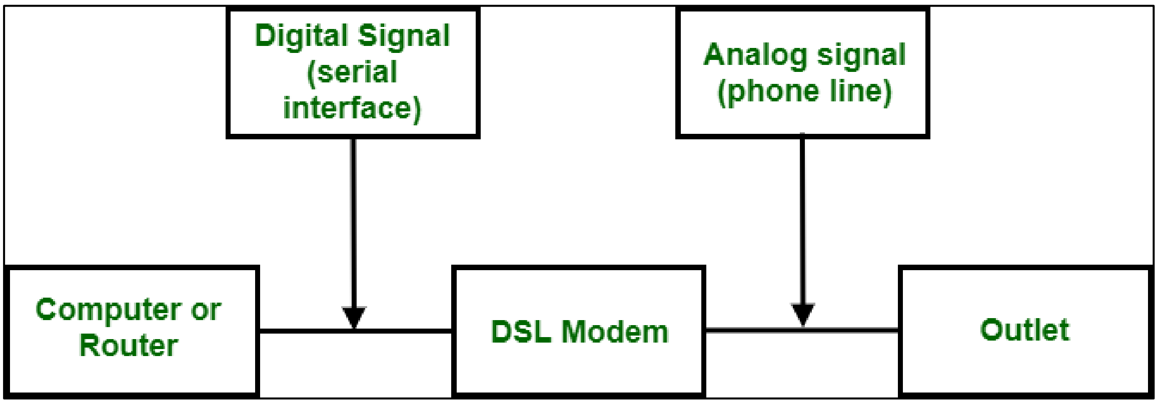
Difference between Synchronous and Asynchronous Transmission

S. No.	Synchronous Transmission	Asynchronous Transmission
1.	In Synchronous transmission, data is sent in form of blocks or frames.	In Asynchronous transmission, data is sent in form of bytes or characters.
2.	Synchronous transmission is fast.	Asynchronous transmission is slow.

S. No.	Synchronous Transmission	Asynchronous Transmission
3.	Synchronous transmission is costly.	Asynchronous transmission is economical.
4.	In Synchronous transmission, the time interval of transmission is constant.	In Asynchronous transmission, the time interval of transmission is not constant, it is random.
5.	In this transmission, users have to wait till the transmission is complete before getting a response back from the server.	Here, users do not have to wait for the completion of transmission in order to get a response from the server.
6.	In Synchronous transmission, there is no gap present between data.	In Asynchronous transmission, there is a gap present between data.
7.	Efficient use of transmission lines is done in synchronous transmission.	While in Asynchronous transmission, the transmission line remains empty during a gap in character transmission.
8.	The start and stop bits are not used in transmitting data.	The start and stop bits are used in transmitting data that imposes extra overhead.
9.	Synchronous transmission needs precisely synchronized clocks for the information of new bytes.	Asynchronous transmission does not need synchronized clocks as parity bit is used in this transmission for information of new bytes.
10.	Errors are detected and corrected in real time.	Errors are detected and corrected when the data is received.
11.	Low latency due to real-time communication.	High latency due to processing time and waiting for data to become available.
12.	Examples: Telephonic conversations, Video conferencing, Online gaming.	Examples: Email, File transfer, Online forms.

xDSL(DSL):

Digital Subscriber Line (DSL) is the most promising for the support of high-speed internet connection over the existing local loops. Basically, it is a set of technologies. **This set is also referred to as xDSL**, where x can be replaced by A, V, H, or S.



Digital Subscriber Line

Technologies of Digital Subscriber Line:

1. ADSL (Asymmetric DSL)
2. ADSL Lite or Universal ADSL or Splitterless ADSL
3. HDSL (High-bit-rate DSL)
4. SDSL (Symmetric DSL)
5. VDSL (Very high bit-rate DSL)

Summary: The following is summary table for technologies of DSL. The values in the following table are approximate and can vary from one implementation to another.

Technology	**Downstream Rate**	**Upstream Rate**	**Distance in feet**	**Twisted Pairs**	**Line Code**
ADSL	1.5-6.1 Mbps	16-640 kbps	12,000	1	DMT
ADSL Lite	1.5 Mbps	500 kbps	18,000	1	DMT
HDSL	1.5-2.0 Mbps	1.5-2.0 Mbps	12,000	2	2B1Q
SDSL	768 kbps	768 kbps	12,000	1	2B1Q
VDSL	25-55 Mbps	3,2 Mbps	3000-10,000	1	DMT

Features of DSL:

- It helps to meet the need for networks in different scenarios like DSL, Fiber/Cable, and 3G/4G Dongle.
- It supports ISP service providers, so we can that it is compatible with most of the service providers.

- It also supports high-quality telephone calls.
- There is an available app to manage the modem.

Introduction to wired and wireless LAN

A network consists of two or more computers that are linked in order to share all form of resources including communication. Both LAN and WLAN networks are interrelated and share moreover common characteristics.

Local Area Network (LAN):

A Local Area Network (LAN) is a type of network that connects devices in a small geographic area, such as a home, office, or school. LANs typically use wired connections, such as Ethernet cables, to connect devices to a central hub or switch. This allows devices to share data, resources, and devices such as printers and storage devices.

Advantages of LAN:

- Speed: LANs provide fast data transfer rates, typically 100 Mbps or 1 Gbps, allowing for quick data transfer between devices.
- Security: LANs are generally more secure than WLANs, as they are physically connected and less susceptible to outside interference.
- Cost: LANs can be less expensive to set up and maintain than WLANs, as they use wired connections that are often less expensive than wireless technology.

Disadvantages of LAN:

- Limited Mobility: LANs typically use wired connections, which limits device mobility and flexibility.
- Limited Range: The range of LANs is limited by the length of the Ethernet cable, which means that devices must be physically located within a small area.
- Installation: The installation of LANs can be complex, requiring the routing of cables and installation of switches.

Wireless Local Area Network (WLAN):

A Wireless Local Area Network (WLAN) is a type of network that uses wireless technology, such as Wi-Fi, to connect devices in the same area. WLANs use wireless access points to transmit data between devices, allowing for greater mobility and flexibility.

Advantages of WLAN:

- Mobility: WLANs provide greater device mobility and flexibility, as devices can connect wirelessly from anywhere within the network range.
- Easy Installation: WLANs are easier to install than LANs, as they do not require physical cabling and switches.
- Range: WLANs can cover a larger area than LANs, allowing for greater device connectivity and flexibility.

Disadvantages of WLAN:

- Security: WLANs are less secure than LANs, as wireless signals can be intercepted by unauthorized users and devices.
- Speed: WLANs provide slower data transfer rates than LANs, typically around 54 Mbps, which can result in slower data transfer between devices.

- Interference: WLANs are susceptible to interference from other wireless devices, which can cause connectivity issues.

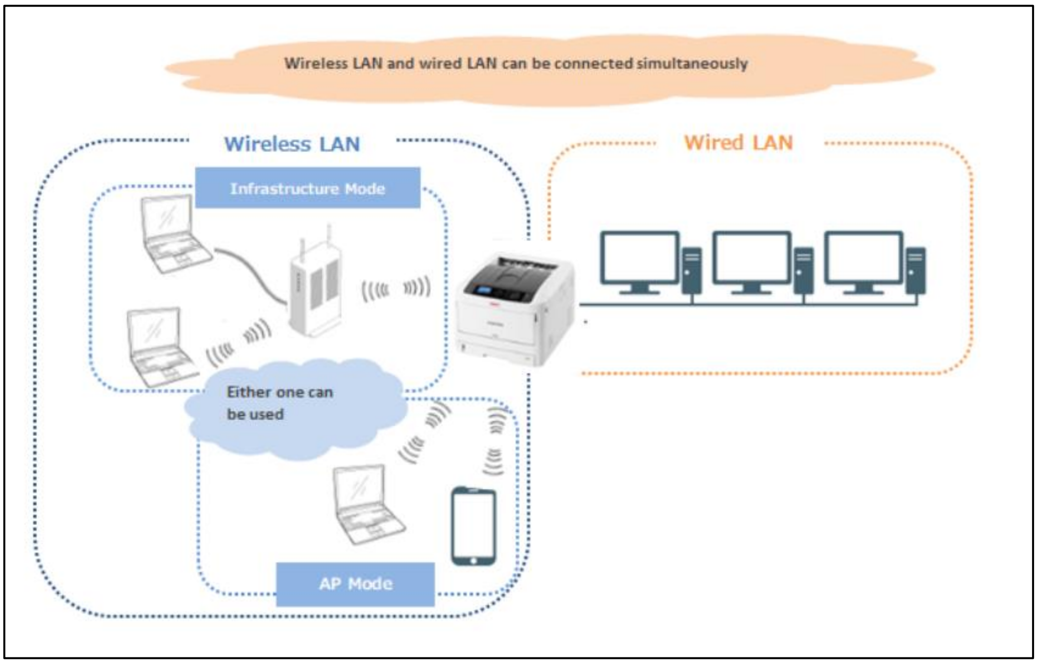
Similarities between LAN and WLAN:

- Both provide connectivity: The primary purpose of both LAN and WLAN is to provide connectivity between devices, allowing them to share data and resources.
- Both use the same protocols: LANs and WLANs use the same protocols for data transfer, such as TCP/IP and Ethernet, which ensures compatibility between devices.
- Both can support multiple devices: Both LANs and WLANs can support multiple devices simultaneously, allowing multiple users to share data and resources.
- Both can be secured: Both LANs and WLANs can be secured using encryption and authentication methods, ensuring that only authorized users have access to the network.
- Both require network hardware: Both LANs and WLANs require network hardware, such as routers, switches, and access points, to function properly.
- Both can be used for internet connectivity: Both LANs and WLANs can be used to connect to the internet, providing access to online resources and services.

Let's discuss about LAN and WLAN:

LAN	WLAN
LAN stands for Local Area Network.	WLAN stands for Wireless Local Area Network.
LAN connections include both wired and wireless connections.	WLAN connections are completely wireless.
LAN network is a collection of computers or other such network devices in a particular location that are connected together by communication elements or network elements.	WLAN network is a collection of computers or other such network devices in a particular location that are connected together wirelessly by communication elements or network elements.
LAN is free from external attacks like interruption of signals, cyber criminal attacks and so on.	Whereas, WLAN is vulnerable to external attacks.
LAN is secure.	WLAN is not secure.
LAN network has lost its popularity due to the arrival of latest wireless networks.	WLAN is popular.

LAN	WLAN
Wired LAN needs physical access like connecting the wires to the switches or routers.	Work on connecting wires to the switches and routers are neglected.
In LAN, devices are connected locally with Ethernet cable.	For WLAN Ethernet cable is not necessary.
Mobility limited.	Outstanding mobility.
It may or may not vary with external factors like environment and quality of cables.	It varies due to external factors like environment and quality of cables. Most of the external factors affect the signal transmission.
LAN is less expensive.	WLAN is more expensive.
Example: Computers connected in a college.	Example: Laptops, cell phones, tablets connected to a wireless router or hotspot.



Both LANs and WLANs have their advantages and disadvantages, depending on the specific requirements. LANs are generally faster and more secure, while WLANs provide greater mobility and flexibility. Choosing the right network for your needs depends on your specific requirements, such as speed, security, and device mobility.